

Real-Time Weather Data Analytics Pipeline

| Automated Data Collection & Dashboard Project

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Automated System for Real-Time Weather Updates

Project Goal

Create a system that **automatically** collects and shows weather data without any manual work.

Problem Solved

No more manual data entry. The system works **24/7 on its own**, saving time and effort.

Main Result

A live Power BI dashboard that updates with **one simple click** to show the latest weather.

Location

Currently tracking **Noida** weather, but the system can easily work for any city in the world.

Simple & Powerful Tools Used



Python

"The Brain"

Fetches weather data from the internet automatically.



MySQL

"The Storage"

Saves all weather history safely in one place.



Power BI

"The Display"

Shows beautiful charts and easy-to-read trends.



GitHub

"The Home"

Keeps our code organized and professional.



Automation

"The Engine"

Keeps the system running non-stop, 24/7.



Weather API

"The Source"

Provides real-time weather info for any city.

How the Data Moves (Step-by-Step)

STEP 1



Fetch

Python asks the OpenWeather API for the latest weather info every few minutes.

STEP 2



Clean

The system removes errors and organizes data into Morning and Night slots.

STEP 3



Save

Clean data is automatically saved into the MySQL database for long-term storage.

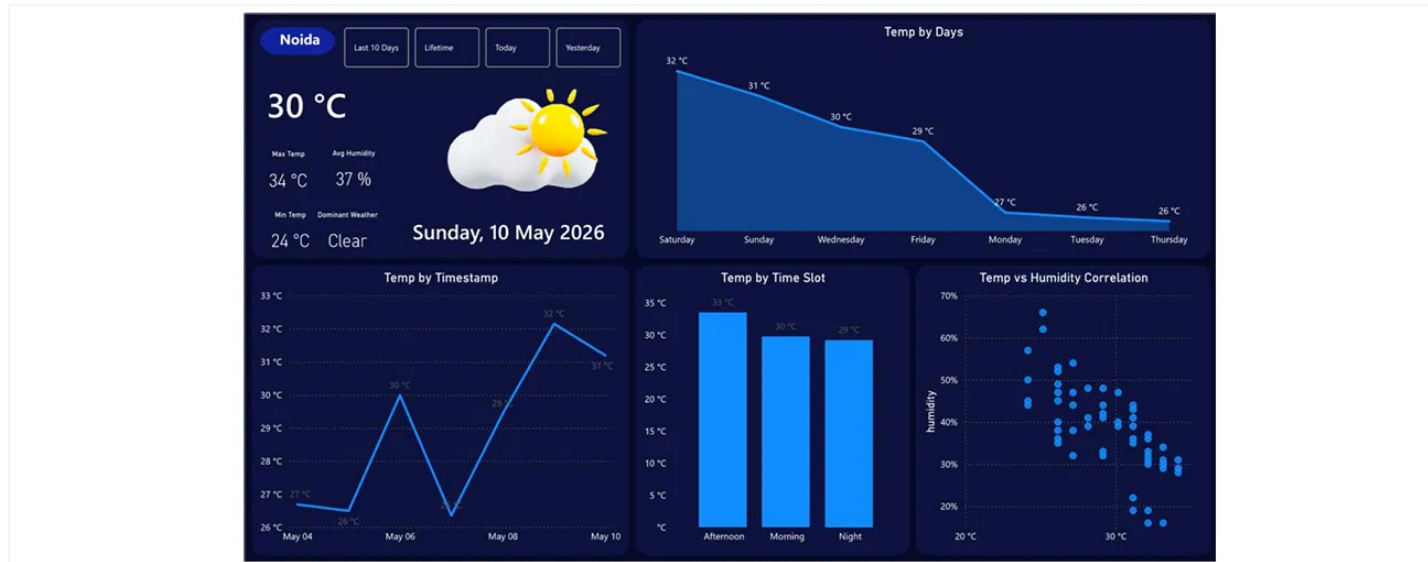
STEP 4



Show

Power BI connects to the database and shows live updates on the dashboard.

Interactive Dashboard: See Everything at Once



Quick View

See **Temperature, Humidity, and Weather** status instantly in one place.

Easy Filters

Simple buttons to see data for **Today, Yesterday, or the Last 10 Days**.

Modern Look

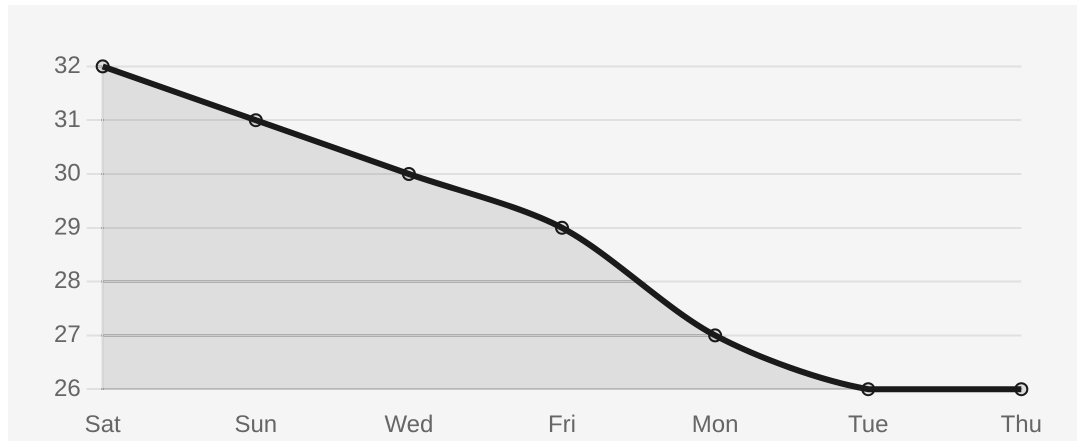
Dark-themed design that is **very easy to read** and understand quickly.

Live Refresh

Just click **one button** to see the most recent weather data updates.

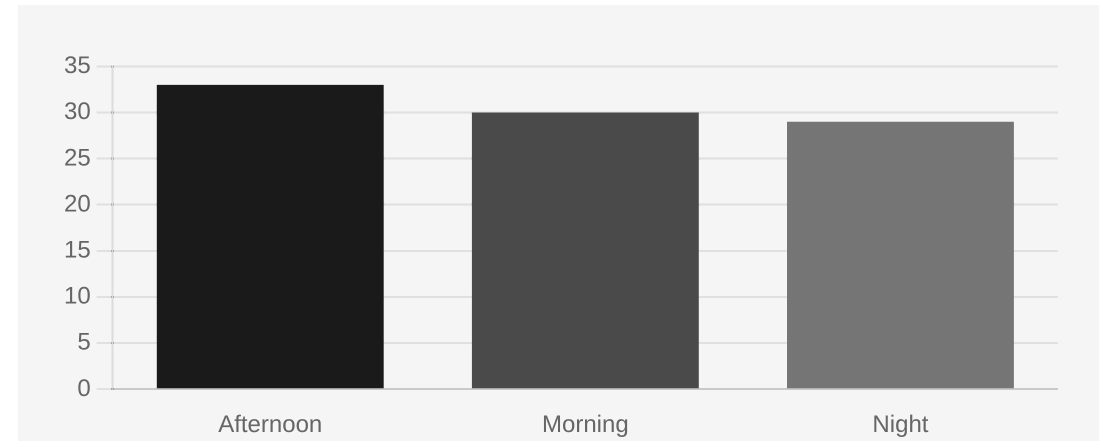
Temperature Trends: Understanding the Day

| WEEKLY VIEW



- See how temperature goes up and down during the week (e.g., 32°C to 26°C).
- Helps us understand when the weather changes the most each day.

| TIME SLOTS



- The system shows that Afternoons are usually the hottest part of the day (33°C avg).
- Smart data that is very useful for planning outdoor work or activities.

Temp vs Humidity: The Hidden Connection

■ The Simple Rule

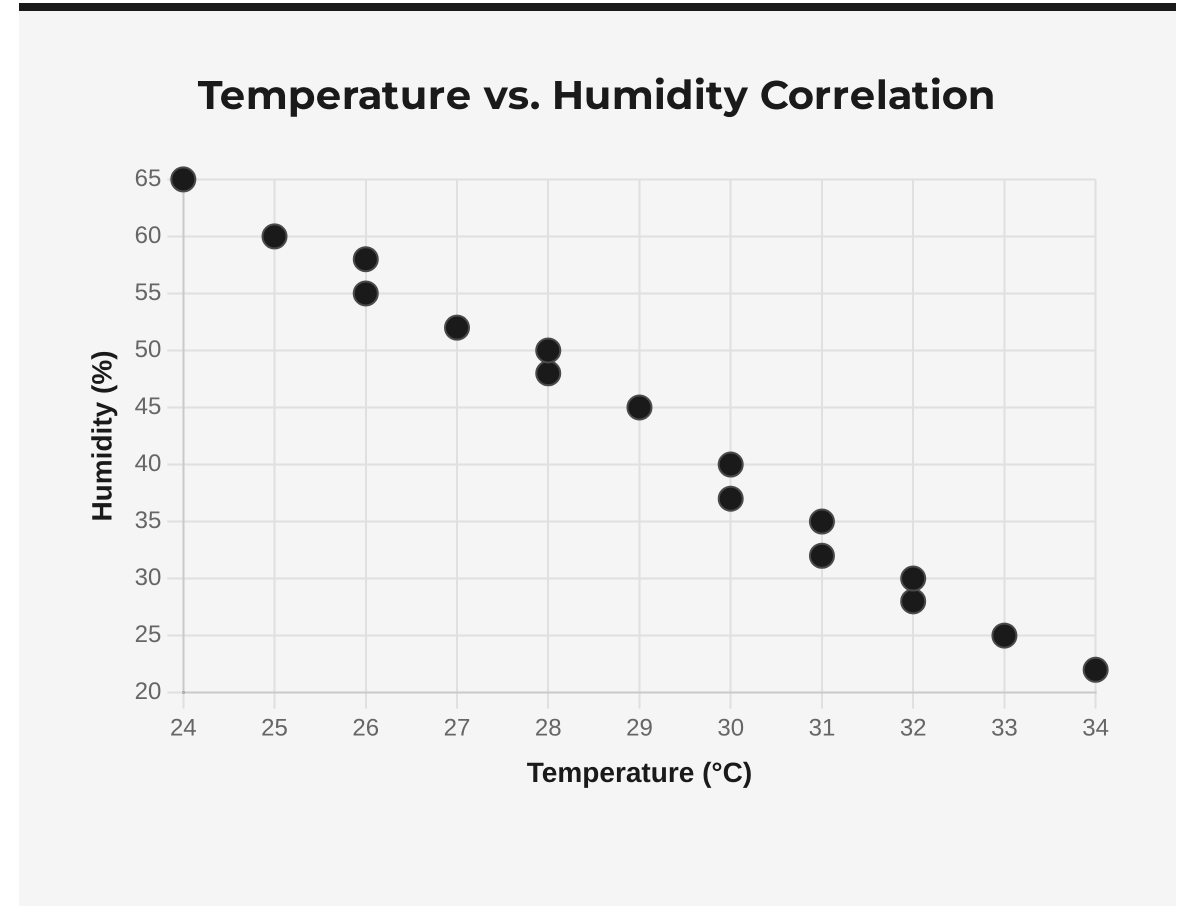
When Temperature goes **UP**, Humidity goes **DOWN**. This is a natural weather pattern.

■ Visual Proof

Our chart clearly shows this "Inverse" relationship. Hot days (30°C+) usually mean very dry air (20-30% humidity).

■ Data Accuracy

This clear pattern proves that our automated data collection is working perfectly and accurately.



Professional Code Standards (GitHub Insights)



Error Handling

The script is **smart**. It doesn't crash if the internet or database fails; it just waits and tries again.



Easy Setup

A simple **requirements file** allows anyone to setup and run the project with just one command.



Logging System

Every event is recorded in a **log file**. This makes it very easy to track what the system is doing.



Fast Database

Used **Primary Keys and Indexing** to keep the database fast, even with thousands of records.

Future Plans: Making it Even Better



More Cities

Add support to track weather for **many cities** at the same time across the world.



Weather Alerts

Send automatic **Email or SMS** alerts for extreme weather like heavy rain or heat.



Cloud Storage

Move the database to **AWS or Azure** so it can be accessed from anywhere easily.



AI Forecasting

Use **AI and Machine Learning** to predict the weather for the next 7 days.

Project Impact: Why it Matters

| Automation & Accuracy

100%

Automated System

No more manual data entry. The system handles everything on its own 24/7.

Real-Time

High Accuracy

Live data from the API is much more accurate and faster than manual reports.

| Professional Quality

Proven Skills

Demonstrates strong expertise in Python, SQL, and Data Analytics for real-world use.

Industry Standard

Uses professional tools like GitHub and clean code that is ready for a work environment.

Thank You!

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